

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418067
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Lee; Seung Hun et al.

Battery pack and method of assembling the same

Abstract

A battery pack, may include: a plurality of cell stacks including a plurality of battery cells; a pack housing accommodating the plurality of cell stacks; a fixing bracket disposed between the plurality of cell stacks and the pack housing and between the plurality of cell stacks; and a fastening member fastening the fixing bracket to the pack housing, wherein the fixing bracket may be disposed to contact the plurality of battery cells on an upper side of the plurality of cell stacks.

Inventors:	Lee; Seung Hun (Daejeon, KR), Rhee; Seo Roh (Daejeon, KR)
Applicant:	SK On Co., Ltd. (Seoul, KR)
Family ID:	1000008818199
Assignee:	SK On Co., Ltd. (Seoul, KR)
Appl. No.:	18/761336
Filed:	July 02, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240356129 A1	Oct. 24, 2024

Foreign Application Priority Data

KR	10-2022-0159270	Nov. 24, 2022
----	-----------------	---------------

Related U.S. Application Data

continuation parent-doc US 18363743 20230802 US 12074334 child-doc US 18761336

Publication Classification

Int. Cl.: H01M50/209 (20210101); H01M10/613 (20140101); H01M10/647 (20140101);
H01M10/6554 (20140101); H01M50/244 (20210101); H01M50/264 (20210101);
H01M50/507 (20210101)

U.S. Cl.:

CPC H01M50/209 (20210101); H01M10/613 (20150401); H01M10/647 (20150401);
H01M10/6554 (20150401); H01M50/244 (20210101); H01M50/264 (20210101);
H01M50/507 (20210101);

Field of Classification Search

CPC: H01M (10/613); H01M (10/647); H01M (10/6554); H01M (50/264); H01M (50/244);
H01M (50/507)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2013/0252058	12/2012	Kim	N/A	N/A
2015/0064541	12/2014	Noh et al.	N/A	N/A
2017/0062789	12/2016	Sim et al.	N/A	N/A
2020/0212390	12/2019	Kume	N/A	H01M 10/6554
2020/0243928	12/2019	Schmidt	N/A	N/A
2020/0411817	12/2019	You et al.	N/A	N/A
2022/0367952	12/2021	Hwang et al.	N/A	N/A
2023/0022636	12/2022	Schmitt et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111490195	12/2021	CN	N/A
111033796	12/2021	CN	N/A
115347298	12/2021	CN	N/A
6680145	12/2019	JP	N/A
2021-007090	12/2020	JP	N/A
10-2013-0107790	12/2012	KR	N/A
10-2015-0024724	12/2014	KR	N/A
10-2017-0025074	12/2016	KR	N/A

OTHER PUBLICATIONS

Extended European Search Report for the European Patent Application No. 23189370.2 issued by the European Patent Office on Feb. 27, 2024. cited by applicant

Primary Examiner: Apicella; Karie O'Neill

Attorney, Agent or Firm: IP & T GROUP LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 18/363,743 filed on Aug. 2, 2023, which claims benefit of priority to Korean Patent Application No. 10-2022-0159270 filed on Nov. 24, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

(1) The present disclosure relates to a battery pack and a method of assembling the same.

2. Description of Related Art

(2) A secondary battery refers to a battery capable of repeated charging and discharging because mutual conversion between chemical energy and electrical energy is reversible.

(3) Such a secondary battery may be used as an energy source for electric vehicles, hybrid vehicles, energy storage systems (ESSs), and the like, which have recently attracted attention, including use in mobile devices.

(4) The secondary battery may be used in a form in which one or more battery cells made of a flexible pouch-type battery cell or a prismatic or cylindrical can-type battery cell having rigidity are electrically connected, and in particular, in the case of an electric vehicle requiring high-output power characteristics, it may be used in a form of a battery module in which one or more cell stacks (in which a plurality of battery cells are stacked) are electrically connected or a battery pack in which one or more battery modules are electrically connected.

(5) Meanwhile, in general, a battery pack includes a housing accommodating one or more battery modules, and the battery module includes a housing accommodating one or more battery cells, separately from the housing of the battery pack. Accordingly, conventional battery packs have redundant components performing the same function, which is disadvantageous in terms of productivity and weight reduction of the battery pack.

SUMMARY

(6) One aspect of the present disclosure is to provide a battery pack whose configuration is simplified as a whole by omitting redundant components even though the same function is performed, and a method of assembling the same.

(7) According to one aspect of the present disclosure, a battery pack may include: a plurality of cell stacks including a plurality of battery cells; a pack housing accommodating the plurality of cell stacks; at least one fixing bracket disposed both between the plurality of cell stacks and the pack housing and between two adjacent cell stacks of the plurality of cell stacks; and a fastening member fastening the fixing bracket to the pack housing, wherein the at least one fixing bracket may be disposed to contact the plurality of battery cells on an upper side of the plurality of cell stacks.

(8) According to one aspect of the present disclosure, the at least one fixing bracket may include: a first fixing bracket disposed to overlap one of the plurality of cell stacks; and a second fixing bracket disposed to overlap the two adjacent cell stacks.

(9) According to one aspect of the present disclosure, the first fixing bracket and the second fixing bracket may have a length of the plurality of battery cells in a stacking direction, and each of the first fixing bracket and the second fixing bracket may include a fastening portion through which the fastening member passes and an extension portion extending from the fastening portion to cover the upper side of the plurality of battery cells.

(10) According to one aspect of the present disclosure, the first fixing bracket may include only one extension portion, and the second fixing bracket may include two extension portions.

(11) According to one aspect of the present disclosure, the battery pack may further include a

support member disposed between the plurality of battery cells and the housing to cover both sides of the plurality of battery cells.

(12) According to one aspect of the present disclosure, one surface of the support member facing the pack housing may include a plurality of protrusions and a plurality of insertion grooves, alternately formed, and the plurality of insertion grooves may be disposed between the plurality of protrusions.

(13) According to one aspect of the present disclosure, one surface of the support member facing the pack housing may be disposed in an inclined shape so that a thickness decreases from an upper side to a lower side of the support member, and the pack housing may be formed in a shape corresponding thereto.

(14) According to one aspect of the present disclosure, the battery pack may further include a compression member disposed between at least two of the plurality of battery cells.

(15) According to one aspect of the present disclosure, the plurality of battery cells comprising the plurality of cell stacks may be prismatic battery cells including an electrode terminal on one side of the plurality of battery cells.

(16) According to one aspect of the present disclosure, the battery pack may further include a bus bar a) a) disposed on a side of one surface of the plurality of battery cells provided with the electrode terminal and b) electrically connected to the electrode terminal; and a bus bar assembly including a bus bar frame on which the bus bar is disposed.

(17) According to one aspect of the present disclosure, the battery pack may further include a cooling plate disposed to face the bus bar assembly and under the plurality of cell stacks.

(18) According to one aspect of the present disclosure, the battery pack may further include a pack cover coupled to the pack housing and covering the plurality of cell stacks.

(19) According to one aspect of the present disclosure, a method of assembling a battery pack may include: a first operation of transferring a plurality of cell stacks including a plurality of battery cells stacked in one direction and a support member disposed to cover the plurality of battery cells in a stacking direction of the plurality of battery cells to a pack housing; a second operation of disposing a fixing bracket between the plurality of cell stacks and the pack housing and between the plurality of cell stacks; and a third operation of fixing the plurality of cell stacks to the pack housing by fastening a fastening member to penetrate through the fixing bracket and the pack housing, wherein in the second operation, the fixing bracket may be disposed to contact the plurality of battery cells on an upper side of the plurality of cell stacks.

(20) According to one aspect of the present disclosure, the plurality of cell stacks may be transferred to the pack housing by a lifting device, and the support member may include a plurality of insertion grooves into which legs of the lifting device are inserted.

(21) According to one aspect of the present disclosure, a compression member may be disposed between at least the plurality of battery cells, to absorb pressure applied in a direction in which the plurality of battery cells are stacked when the plurality of cell stacks are transferred.

(22) According to still another aspect of the present disclosure, a battery pack may include: a plurality of cell stacks including a plurality of battery cells; a pack housing for accommodating the plurality of cell stacks; at least one fixing bracket configured to fix the plurality of cell stacks to opposite sidewalls of the pack housing; a fastening member fastening the at least one fixing bracket to the housing; and a support disposed on both sides in the arrangement direction of the plurality of battery cells, one surface of the support member may face the inner surface of the pack housing, and the surface of the support member and the pack housing facing each other may have an inclined shape.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is an exploded perspective view of a battery pack according to one embodiment of the present disclosure.
- (3) FIG. 2 is a plan view illustrating an assembled state of a battery pack according to one embodiment of the present disclosure.
- (4) FIGS. 3 and 4 are side views illustrating an assembled state of a battery pack according to one embodiment of the present disclosure.
- (5) FIGS. 5 to 7 are views illustrating a cell stack assembled into a battery pack according to one embodiment of the present disclosure.
- (6) FIG. 8 is a view illustrating a process of assembling a cell stack according to one embodiment of the present disclosure into a battery pack.
- (7) FIGS. 9 and 10 are diagrams illustrating a battery pack to which a cooling structure according to various embodiments of the present disclosure is applied.

DETAILED DESCRIPTION

- (8) Prior to the detailed description of the present disclosure, the terms or words used in the present specification and claims described below should not be construed as being only understood from a common or dictionary meaning, as the inventor(s) may have described below in certain instances his/her invention using their own terminology. Additionally, the terms may be understood in view of the technical content of the present disclosure. Moreover, the embodiments described in this specification and the configurations shown in the drawings are illustrative embodiments of the present disclosure, and there may be various equivalents and modifications that can be substituted therefor.
- (9) Hereinafter, various embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. At this time, it should be noted that the same components in the accompanying drawings are indicated by the same reference numerals as much as possible. In addition, detailed descriptions of well-known functions and configurations will be omitted below, and for the same reason, some components in the accompanying drawings are exaggerated, omitted, or schematically illustrated, and the size of each component does not necessarily reflect the actual size thereof.
- (10) In the following description, singular expressions include plural expressions unless the context clearly dictates otherwise. The terms “include” or “comprise” are intended to indicate that there is a feature, number, step, operation, component, part, or combination thereof described in the specification, but it should be understood that the presence or addition of one or more other features or numbers, steps, operations, components, parts, or combinations thereof is not precluded.
- (11) In addition, in the following description, expressions such as “above,” “upper,” “below,” “lower,” “front,” “rear,” and the like, are expressed based on the directions illustrated in the drawings, and may be expressed differently if a direction of the object is changed.
- (12) In addition, terms including ordinal numbers such as “first” and “second” used herein may be used to describe various components, but the components are not limited by the terms, and the terms are used only for the purpose of distinguishing one component from another component. For example, without departing from the scope of the present invention, a first element may be termed a second element, and similarly, a second element may be termed a first element.
- (13) FIG. 1 is an exploded perspective view of a battery pack according to one embodiment of the present disclosure.
- (14) Referring to FIG. 1, a battery pack **1000** according to one embodiment of the present disclosure may include a pack housing **1010**, a plurality of cell stacks (or battery module) **1100**

accommodated in the pack housing **1010**, and a pack cover **1020** coupled to the pack housing **1010** to cover the plurality of cell stacks **1100**. For example, the pack housing **1010** and the pack cover **1020** forming an exterior of the battery pack **1000** may be formed of a material containing aluminum.

(15) The pack housing **1010** may form an inner space in which a plurality of cell stacks **1100** are accommodated. The inner space of the pack housing **1010** may be partitioned into a plurality of spaces so that the plurality of cell stacks **1100** can be individually disposed.

(16) A sidewall **1011** may be formed in the pack housing **1010** to penetrate through an inner space, and the inner space may be partitioned or divided into a plurality of spaces by the sidewall **1011**. Each of the cell stacks **1100** may be individually disposed in one of the spaces, partitioned by the sidewall **1011**.

(17) In one embodiment, referring to the accompanying drawings, the sidewall **1011** may have a shape extending in a Y-direction, and as the sidewall **1011** is disposed, the inner space may be partitioned in an X-direction. The plurality of cell stacks **1100** may be disposed in respective spaces, partitioned in the X-direction.

(18) A pack cover **1020** may be coupled to an upper portion of the pack housing **1010** in a state in which the plurality of cell stacks **1100** are disposed in the inner space. The pack cover **1020** may be coupled to the pack housing **1010** to completely cover the inner space. The plurality of cell stacks **1100** disposed in the inner space may be protected from external impacts by the pack cover **1020**.

(19) According to one embodiment of the present disclosure, the plurality of cell stacks **1100** may include a plurality of battery cells **1110** arranged in one direction. In one embodiment, referring to the drawings, the plurality of cell stacks **1100** may include a plurality of battery cells **1110** arranged in a Y-direction. A support member **1140** supporting the plurality of battery cells **1110** may be disposed outside the plurality of battery cells **1110**. The support member **1140** may support the plurality of battery cells **1110** in the Y-direction in which the plurality of battery cells **1110** are arranged. A description related to the support member **1140** will be described later.

(20) According to one embodiment of the present disclosure, the plurality of battery cells **1110** may be prismatic battery cells **1110**. That is, the battery module **1100** may include a plurality of prismatic battery cells **1110** arranged in a Y-direction. The plurality of prismatic battery cells **1110** may be arranged in the Y-direction so that wide surfaces thereof face each other.

(21) The prismatic battery cell may refer to a battery cell in which a case **1111** accommodating an electrode assembly and an electrolyte has a flat and angular shape, and the case **1111** may be made of aluminum.

(22) According to one embodiment of the present disclosure, the plurality of prismatic battery cells **1110** may have positive and negative electrode terminals (or electrode terminals) protruding to one side thereof for example in a height direction (e.g., Z-direction). The positive electrode terminal **1112** and the negative electrode terminal **1113** may be provided to be spaced apart from each other in a length direction (e.g., X-direction) of the battery cell **1110**. The plurality of battery cells **1110** may be electrically connected to neighboring battery cells **1110** through an electrode terminal.

(23) In addition, referring to the drawings, a vent portion **1115** (shown in FIG. 2) may be provided in the case **1111** between the positive electrode terminal **1112** and the negative electrode terminal **1113**. The vent portion **115** may be a passage through which gas generated inside the case **1111** is discharged.

(24) According to one embodiment of the present disclosure, a bus bar assembly **1200** (as shown in FIG. 1) may be disposed on an upper side of the plurality of cell stacks **1100**. The bus bar assembly **1200** may (as shown in FIG. 1) include a bus bar **1210** electrically connected to an electrode terminal and a bus bar frame **1220** on which the bus bar **1210** is disposed. In addition, the bus bar assembly **1200** may include a flexible printed circuit board (FPCB) **1230** (as shown in FIG. 1) electrically connected to the bus bar **1210** and receiving a voltage, temperature information, and the like, of the plurality of battery cells **1110**.

(25) According to one embodiment of the present disclosure, the bus bar frame **1220** may extend in a longitudinal direction (Y-direction based on the drawing) of the cell stack **1100**, and may overlap the plurality of battery cells **1110** on an upper side of the battery module **1100**.

(26) Meanwhile, a portion of both sides of the plurality of battery cells **1110** in the longitudinal direction (X-direction based on the drawing) may not overlap the bus bar frame **1220** (in this case, the positive electrode terminal **1112** and the negative electrode terminal **1113** of at least the plurality of battery cells **1110** may overlap the bus bar frame **1220**), and the corresponding portion may overlap a fixing bracket **1300** to be described later. That is, a bus bar assembly **1200** and a fixing bracket **1300** may be disposed on an upper side of the cell stack **1100** according to one embodiment of the present disclosure.

(27) The bus bar frame **1220** may include a hole exposing electrode terminals **1112** and **1113** of the plurality of battery cells **1110** and vent portion **1115**, and the bus bar **1210** may be electrically connected to the electrode terminals **1112** and **1113** exposed through the hole.

(28) The bus bar **1210** may electrically connect electrode terminals **1112** and **1113** of neighboring battery cells **1110**. In the accompanying drawings, a plurality of battery cells **1110** as illustrated are connected in series through the bus bar, but this embodiment thereof is not limited thereto, and various connection structures may be employed as needed.

(29) According to one embodiment of the present disclosure, a fixing bracket **1300** may be disposed on an upper side of the plurality of cell stacks **1100**. The fixing bracket **1300** may be a component fixing and coupling the cell stack **1100** to the battery pack **1000**.

(30) FIG. 2 is a plan view illustrating an assembled state of the battery pack according to one embodiment of the present disclosure, and FIGS. 3 and 4 are side views illustrating an assembled state of the battery pack according to one embodiment of the present disclosure.

(31) The cell stack **1100** according to one embodiment of the present disclosure may not include a housing or top cover closing the plurality of battery cells **1110** except for the support member **1140**. Accordingly, the plurality of battery cells **1110** may be directly fixed and coupled to the pack housing **1010**.

(32) The fixing bracket **1300** may be disposed to simultaneously overlap the plurality of cell stacks **1100** and the pack housing **1010** (including the side walls **1011**). The fixing bracket **1300** may have a length in an X-direction so as to cover the cell stack **1100** and the pack housing **1010** at the same time, and may have a length in a Y-direction so as to overlap the plurality of battery cells **1110** included in the cell stack **1100** at the same time.

(33) Both sides of the plurality of cell stacks **1100** may be supported by the fixing brackets **1300**. Referring to FIGS. 2 and 3, the fixing bracket **1300** may be disposed between the cell stack **1100** and the pack housing **1010** and between the plurality of cell stacks **1100**. While the fixing bracket **1300** illustrated in FIGS. 1 and 2 comprises separate elements **1310**, **1320**, these elements **1310** and **1320** may be connected together or otherwise comprise a unitary piece.

(34) The fixing bracket **1300** may include a first fixing bracket **1310** and a second fixing bracket **1320**. The first fixing bracket **1310** may be a fixing bracket disposed between the cell stack **1100** and the pack housing **1010**, and the second fixing bracket **1320** may be a fixing bracket disposed between the plurality of cell stacks **1100**.

(35) The fixing bracket **1300** may include two first fixing brackets **1310** and one or more second fixing brackets **1320**. Since the first fixing bracket **1310** is disposed between the cell stack **1100** and the pack housing **1010**, the two first fixing brackets **1310** may be provided. Meanwhile, a different number of second fixing brackets **1320** disposed between the plurality of cell stacks **1100** may be provided according to the number of the plurality of cell stacks **1100**. For example, when the battery pack **1000** includes n cell stacks **1100**, n-1 second fixing brackets **1320** may be provided.

(36) The first fixing bracket **1310** and the second fixing bracket **1320** may include extension portions **1312** and **1322** disposed on an upper side of the cell stack **1100**.

(37) In detail, the first fixing bracket **1310** and the second fixing bracket **1320** may include fastening portions **1311** and **1321** disposed on the upper side of the pack housing **1010** or the side wall **1011**, and extension portions **1312** and **1322** disposed to overlap the upper side of the cell stack **1100**.

(38) The fastening portions **1311** and **1321** may have a thickness to compensate for a height difference between the pack housing **1010** or the side wall **1011** and the cell stack **1100**. In addition, the fastening portions **1311** and **1321** may be formed with a plurality of through-holes **1313** and **1323** into which a fastening member **1400** is inserted in a longitudinal direction.

(39) As illustrated in FIG. 3, the fastening member **1400** may be fastened to the pack housing **1010** in the z direction. In this case, multiple fastening members **1400** are inserted into the through-holes **1313** and **1323** in fixing brackets **1310** and **1320**.

(40) The side wall **1011** of the pack housing **1010** may include a fastening hole **1012** (shown in FIG. 1) in which the fastening member **1400** is accommodated and fastened, and the fastening member **1400** may be fastened to the pack housing **1010** while being inserted into the fastening hole **1012**.

(41) The extension portions **1312** and **1322** may be portions extending from the fastening portions **1311** and **1321** in the X-direction. The first fixing bracket **1310** and the second fixing bracket **1320** may be disposed so that the extension portions **1312** and **1322** are placed toward the battery module **1100**, and the extension portions **1312** and **1322** may be disposed to overlap a portion of the cell stacks **1100** on the upper side of the cell stack **1100**. The plurality of cell stacks **1100** may be pressed and fixed in the z direction by the extension portions **1312** and **1322** in a state in which the fastening member **1400** is fastened to the fastening portions **1311** and **1321**.

(42) Since the first fixing bracket **1310** is disposed between the cell stack **1100** and the pack housing **1010**, the first fixing bracket **1310** may include only one extension portion **1311**. In addition, since the second fixing bracket **1320** is disposed between the plurality of cell stacks **1100**, the second fixing bracket **1320** may include two extension portions **1321**. For example, a cross-section of the first fixing bracket **1310** may be a 'custom character' or 'custom character' shape, and a cross-section of the second fixing bracket **1320** may be a 'custom character' shape.

(43) Meanwhile, FIG. 4 is a view illustrating a state in which the cell stack **1100** is assembled to the pack housing **1010** when viewed from the X-direction. Prior to this description, the cell stack **1100** according to one embodiment of the present disclosure will be described with reference to FIGS. 5 to 7.

(44) FIGS. 5 and 7 are views illustrating a cells stack assembled into a battery pack according to one embodiment of the present disclosure.

(45) A cell stack **1100** according to this embodiment of the present disclosure may include a plurality of prismatic battery cells **1110** arranged in one direction. For example, the plurality of prismatic battery cells **1110** may be arranged in a Y-direction so that wide surfaces thereof face each other.

(46) The plurality of battery cells **1110** may be arranged in the Y-direction at a predetermined interval, and a compression member **1150** may be disposed at intervals between each of the plurality of battery cells **1110**. The compression member **1150** may be disposed between the plurality of battery cells **1110** to absorb swelling of the battery cells **1110** and maintain the interval between the plurality of battery cells **1110**. For example, the compression member **1150** may be formed of a material such as polyurethane. In addition, as will be described later, the compression member **1150** may prevent the plurality of battery cells **1110** from sagging when the battery module **1100** is lifted.

(47) The cell stack **1100** according to one embodiment of the present disclosure may include a support member **1140** on both sides of an arrangement direction of the plurality of battery cells **1110**. The support member **1140** may be disposed outside the arrangement direction of the plurality of battery cells **1110** to support the plurality of battery cells **1110** in the y-direction. The support

member **1140** may be fixed to a side surface of the battery cell **1110**, disposed on an outermost side thereof or a compression member **1150**, through an adhesive material. The support member **1140** may reduce a swelling phenomenon of the plurality of battery cells **1110** according to charging and discharging.

(48) The support member **1140** may be formed of a plastic material, for example, a material such as polypropylene, and more specifically, may be provided in a form of a plastic injection molding into which die cast aluminum (Al) is inserted to reinforce rigidity. In addition, the support member **1140** may include a bus bar **1210**, in detail, a part to which a HV terminal of the bus bar is connected.

(49) Referring to FIG. 7, the support member **1140** may include an insertion groove **1141** on one surface facing the pack housing **1010**. The insertion groove **1141** may be formed in a height direction (e.g., z direction based on the drawing), and may be provided in plurality in a width direction of the cell stack **1100**. A protrusion **1142**, protruding relatively more than an insertion groove **1141** may be alternately formed with an insertion groove **1141** on the one surface of the support member **1140**, and the insertion groove **1141** may respectively be formed between the protrusions **1142**. Legs **150** of a device (hereinafter, referred to as a lifting device) for transferring the cell stack **1100** to the pack housing **1010** may be inserted into the insertion groove **1141**.

(50) FIG. 8 is a view illustrating a process of assembling a cell stack according to one embodiment of the present disclosure into a battery pack.

(51) Referring to FIG. 8, a cell stack **1100** according to this embodiment of the present disclosure may be transported by a lifting device **100** and assembled to a pack housing **1010**. Specifically, the lifting device **100** may lift the cell stack **1100** in a state in which the leg **150** of the lifting device **100** is inserted into an insertion groove **1141** of the support member **1140** and transfers the same into the pack housing **1010**. When the cell stack **1100** is lifted in a state in which the leg **150** of the lifting device is inserted into the insertion groove **1141**, since the battery module **1100** does not include a separate housing, a sagging phenomenon of the plurality of battery cells **1110** in a direction of gravity may occur. Accordingly, as described above, the battery module **1100** according to one embodiment of the present disclosure may prevent this by including the compression member **1150** between the plurality of battery cells **1110**.

(52) According to one embodiment of the present disclosure, as illustrated in FIG. 4, the support member **1140** may have a shape in which one surface thereof facing the pack housing **1010** is inclined. For example, the support member **1140** may have a shape in which the thickness decreases from an upper side to a lower side, and from the viewpoint of the cell stack **1100**, may have a shape in which a length of the cell stack **1100** may be reduced. An inner side surface of the pack housing **1010** facing the one surface of the support member **1140** may have an inclined shape corresponding thereto. When the support member **1140** has such a shape, the cell stack **1100** may be more easily assembled to the pack housing **1010** along the inclined surface of the pack housing **1010**. However, this is merely one embodiment of the present disclosure, and the support member **1140** may not have an inclined shape.

(53) Meanwhile, according to another embodiment of the present disclosure, a gap filler (such as gap filler **1510** described below) may be disposed between the pack housing **1010** and the cell stack **1100**. The gap filler has a relatively weak adhesive force, so that the cell stack **1100** may be attached or detached, and has an advantage in that the cell stack **1100** may be easily replaced.

(54) FIGS. 9 and 10 are diagrams illustrating a battery pack to which a cooling structure according to embodiments of the present disclosure is applied.

(55) Referring to FIGS. 9 and 10, in a battery pack **1000** according to one embodiment of the present disclosure, a cooling structure (hereinafter referred to as a cooling plate) may be applied to an upper or a lower side of a pack housing **1010**. In other words, in the battery pack **1000** according to this embodiment of the present disclosure, a cooling plate does not necessarily need to be disposed at a predetermined location, and may be selectively disposed on an upper side or a lower side of the pack housing **1010**.

(56) In one embodiment, the battery pack **1000** may include a cooling plate **1500** between a bottom surface of the pack housing **1010** and a cell stack **1100**. In this case, the cell stack **1100** may be disposed so that an electrode terminal of the plurality of battery cells **1110** face an upper portion of the battery pack **1000**, that is, a pack cover **1020**. In another embodiment, the battery pack **1000** may include a cooling plate **1500** between the pack cover **1020** and the cell stack **1100**. In this case, the cell stack **1100** may be disposed so that an electrode terminal of the plurality of battery cells **1110** face a lower portion of the battery pack **1000**, that is, the bottom surface of the pack housing **1010**.

(57) Meanwhile, a gap filler **1510** may be further disposed between the cooling plate **1500** and the cell stack **1100**. For example, the gap filler **1510** may be a thermal pad that facilitates thermal transfer. Accordingly, when the gap filler **1510** is provided such as a thermal pad, cooling efficiency may be increased. Alternately, a gap filler **1510** may be provided as a thermal interface material (TIM) with non-adhesive for recycling or reusing.

(58) As described above, the battery pack **1000** according to one embodiment of the present disclosure may not include a separate cover covering a plurality of battery cells **1110**, excluding a support member **1140** disposed in a stacking direction of the plurality of battery cells **1110** to transfer the cell stack **1100**, and may replace a cover of the battery module **1100** in which the pack housing **1010** is omitted. Accordingly, economic feasibility of manufacturing the battery pack **1000** may be secured, and sufficient rigidity may be maintained through the thickness of the pack housing **1010**, and the like.

(59) As set forth above, according to one embodiment of the present disclosure, a battery pack has an advantage in that productivity is improved, and it is advantageous for reducing weight.

(60) In addition, the battery pack according to one embodiment of the present disclosure may secure sufficient rigidity even though it has a simplified structure.

(61) The present disclosure provides detail through specific examples given above. The examples given above are only examples to help the understanding of the present disclosure, and the scope of the present disclosure is not limited thereto.

(62) While this disclosure includes specific examples, it will be apparent to one of ordinary skill in the art that various changes in form and details may be made in these examples without departing from the scope of the disclosure and any equivalents to the details provided. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation of the claims. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed to have a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not only by the detailed description but also by any equivalents.

Claims

1. A battery pack, comprising: a plurality of cell assemblies including a plurality of battery cells; a pack housing accommodating the plurality of cell assemblies; at least one fixing bracket disposed both between the plurality of cell assemblies and the pack housing and between two adjacent cell assemblies of the plurality of cell assemblies; and a fastening member fastening the fixing bracket to the pack housing, wherein the at least one fixing bracket is disposed to contact the plurality of battery cells on an upper side of the plurality of cell assemblies, wherein each of the plurality of cell assemblies further comprises a support member disposed between the plurality of battery cells and the pack housing to cover both sides of the plurality of battery cells, wherein one surface of the support member, facing the pack housing, comprises a plurality of protrusions and a plurality of insertion grooves, alternately formed, and the plurality of insertion grooves is disposed between the

plurality of protrusions, wherein the plurality of insertion grooves is recessed from a surface of the support member facing the pack housing toward an opposite surface of the support member facing the plurality of battery cells, and wherein the plurality of protrusions and insertion grooves extend from the upper side to the lower side of the support member along the surface of the support member facing the pack housing.

2. The battery pack of claim 1, wherein the at least one fixing bracket comprises a first fixing bracket disposed to overlap one of the plurality of cell assemblies; and a second fixing bracket disposed to overlap the two adjacent cell assemblies.

3. The battery pack of claim 2, wherein the first fixing bracket and the second fixing bracket have a length of the plurality of battery cells in a stacking direction, and each of the first fixing bracket and the second fixing bracket comprises a fastening portion through which the fastening member passes and an extension portion extending from the fastening portion to cover the upper side of the plurality of battery cells.

4. The battery pack of claim 3, wherein the extension portion of the first fixing bracket comprises only one extension portion, and the extension portion of the second fixing bracket comprises two extension portions.

5. The battery pack of claim 1, wherein one surface of the support member, facing the pack housing, is disposed in an inclined shape so that a thickness decreases from an upper side to a lower side of the support member, and the pack housing is formed in a shape corresponding thereto.

6. The battery pack of claim 1, wherein the plurality of cell assemblies further comprise a compression member disposed between at least two of the plurality of battery cells.

7. The battery pack of claim 1, wherein the plurality of battery cells comprising the plurality of cell assemblies are prismatic battery cells including an electrode terminal on one side of the plurality of battery cells.

8. The battery pack of claim 7, further comprising: a bus bar a) disposed on a side of one surface of the plurality of battery cells provided with the electrode terminal and b) electrically connected to the electrode terminal; and a bus bar assembly including a bus bar frame on which the bus bar is disposed.

9. The battery pack of claim 8, further comprising: a cooling plate disposed opposite to the bus bar assembly and under the plurality of cell assemblies.

10. The battery pack of claim 1, further comprising: a pack cover coupled to the pack housing and covering the plurality of cell assemblies.
